H_∞ Control Design for an Inverted Pendulum System

1 Overview

H_∞ control synthesis offers a robust control framework designed to minimize the worst-case gain from disturbances to the controlled outputs. This method is particularly effective for systems subject to uncertainties and external disturbances, ensuring robust performance even when model mismatches occur.

2 Augmented Plant and Weighting Functions

The H_∞ controller is synthesized based on an augmented (generalized) plant, which is formed by combining the nominal plant with specific weighting functions:

- **Performance Weight $W_p(s)$:**

  $$W_p(s) = \frac{s+10}{s+0.1}$$

  This weight shapes the sensitivity function and emphasizes tracking performance. Increasing the numerator (e.g., modifying $s+10$) imposes a higher penalty on tracking errors at higher frequencies, leading to a more aggressive response. Conversely, changing the denominator (e.g., using $s+0.5$ instead of $s+0.1$) reduces the high-frequency gain, enhancing robustness but potentially slowing down the response.

- **Control Weight W_u:**

  $$W_u = 0.1$$

  This weight penalizes the control effort. A smaller value allows the controller to exert stronger control actions (more aggressive behavior), while a larger W_u results in a more conservative control effort, potentially sacrificing transient performance for improved robustness and reduced actuator saturation.

The augmented plant P is then constructed (using MATLAB's `augw` function) to incorporate these weights into the synthesis process.

3 H$_\infty$ Synthesis Procedure

The H$_\infty$ synthesis process aims to compute a controller K_{hinf} such that the closed-loop system minimizes the worst-case disturbance amplification:

$$\min_K \|T_{zw}(K)\|_\infty < \gamma,$$

where $T_{zw}(K)$ is the transfer function from exogenous disturbances to controlled outputs, and γ is the achieved performance level. The MATLAB function `hinfsyn` is employed to solve this optimization problem, returning the controller K_{hinf}, the closed-loop system CL, the performance level γ, and additional synthesis information.

4 Influence of Tuning Parameters

The weighting functions $W_p(s)$ and W_u play a crucial role in determining the behavior of the H$_\infty$ controller:

- **Performance Weight $W_p(s)$:**
  - *Numerator Adjustment:* Increasing the gain in the numerator (e.g., from $s+10$ to $s+20$) increases the penalty on tracking errors, forcing the controller to achieve tighter tracking performance at the expense of potentially larger control actions.
  - *Denominator Adjustment:* Increasing the denominator term (e.g., from $s+0.1$ to $s+0.5$) reduces the high-frequency gain. This modification improves robustness by limiting noise amplification but might slow the response to rapid changes.
- **Control Weight W_u:**
  - A lower W_u allows more aggressive control inputs, which may enhance transient response and reduce rise time but could lead to actuator saturation or increased sensitivity to unmodeled dynamics.
  - A higher W_u imposes a greater penalty on control effort, promoting a more conservative controller that may exhibit slower response but improved robustness and reduced risk of actuator saturation.

The achieved performance level γ is indicative of how well the closed-loop system attenuates disturbances. A lower γ represents better performance, though it may require a more complex controller.

5 Closed-Loop Analysis and Frequency-Domain Interpretation

After synthesis, the H$_\infty$ controller is integrated with the augmented plant via a Linear Fractional Transformation (LFT) to form the closed-loop system. Key analysis methods include:

- **Eigenvalue Analysis:** Verifying that all closed-loop poles lie in the stable region ensures system stability.
- **Time-Domain Response:** The step response is analyzed to evaluate transient performance metrics such as rise time and settling time.
- **Frequency-Domain Analysis:**
  - *Root Locus:* Demonstrates the variation of closed-loop poles with respect to the control gain.
  - *Nyquist Plot:* Provides insights into the robust stability margins and potential sensitivity to model uncertainties.
  - *Bode Plot:* Shows how the controller attenuates disturbances over a range of frequencies, revealing phase and gain margins.

6 Summary

The H_∞ control strategy emphasizes robust performance by explicitly accounting for worst-case disturbance rejection. By tuning the performance weight $W_p(s)$ and control weight W_u, one can balance aggressive tracking against robustness and control effort. Adjustments in these parameters directly influence the closed-loop performance—both in the time domain (through transient response characteristics) and the frequency domain (through stability margins and disturbance attenuation). The synthesis process ensures that the controller achieves a specified performance level γ, making it a powerful tool for systems requiring high robustness under uncertainty.